

V17.54.3 HUD Safety Hotfix — Fix Report

Status: ⚠️ AWAITING USER AUDIT — DO NOT MERGE
Date: 2026-05-28
Branch: hotfix/hud-tracker-v17.54.3
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Commit Hashes

Ref	Hash	Description
v17.54.2 (last good main)	3124db2	Backend update to v17.54.2
v17.54.3 (original push)	316dd11	Direct push to main (REVERTED)
Revert on main	86dbc46	Revert of 316dd11 — main restored to v17.54.2
Hotfix branch base	316dd11	Branched from original v17.54.3
Hotfix fix commit	43a6f9c	Gap fixes: row map, guards, counters, pdCryptoMode

Branch Status

- **main:** Rolled back to v17.54.2 state (86dbc46)
- **hotfix/hud-tracker-v17.54.3:** Contains all v17.54.3 safety fixes + 3 gap fixes (43a6f9c)
- **Railway deployment:** Running v17.54.2 (main was reverted)

3 Specific Gaps Fixed

Gap 1: maxHudRows — 22 → 21 (Justified with Row Map)

Problem: maxHudRows = 22 was arbitrary padding (1 extra unused row).
Fix: Changed to maxHudRows = 21 — exact fit for 21-row HUD.
Justification: Full HUD Row Map documented in code (lines 749-770):

Row 0:	Header	☐ "SMC P&D" / version badge
Row 1:	Bias	☐ H4 P&D bias (Bullish/Bearish/Neutral)
Row 2:	Align	☐ Trend alignment status
Row 3:	KZ	☐ Kill Zone status
Row 4:	Sniper	☐ Sniper signal progress
Row 5:	POI	☐ Point of Interest status
Row 6:	Momentum	☐ Momentum direction
Row 7:	Guardian	☐ Guardian filter status
Row 8:	Confl	☐ Confluence score
Row 9:	AMD	☐ AMD phase
Row 10:	Trade	☐ Trade context
Row 11:	Entry	☐ Entry mode
Row 12:	Action	☐ Action step + plan
Row 13:	Stage	☐ Signal stage
Row 14:	Alert	☐ Last alert type
Row 15:	Signals	☐ Signals today count
Row 16:	Levels	☐ Session levels
Row 17:	ZoneCfg	☐ [debug] Zone configuration
Row 18:	ZoneHt	☐ [debug] Zone height / cap status
Row 19:	ZoneSt	☐ [debug] Zone boolean states
Row 20:	Ratio	☐ [debug] Zone ratio check

Why not 12: The HUD has 17 standard rows (0-16) + 4 debug rows (17-20) = 21 total. Reducing to 12 would silently drop rows 12-20 (Action, Stage, Alert, Signals, Levels, and all debug rows). These are all existing features from v17.54.2.

Gap 2: f_setCell Guard Pattern — Reordered to Match Spec

Before (v17.54.3 original):

```
if not na(_t) and _col >= 0 and _col < maxHudCols and _row >= 0 and _row < maxHudRows
```

After (hotfix — matches user spec exactly):

```
if not na(_t) and _col >= 0 and _row >= 0 and _col < maxHudCols and _row < maxHudRows
```

Change: Guard order now matches the exact pattern from the hotfix brief: `_col >= 0 and _row >= 0` grouped together before the upper-bound checks.

Note on table.cell vs table.cell_set_text: The wrapper uses `table.cell()` instead of `table.cell_set_text()` because 14+ HUD cells have **dynamic colors** that change per bar (e.g., `h4ZoneColor`, `sniperProgressColor`, `confluenceColor`). Using `cell_set_text` would freeze colors at init time. `table.cell()` handles both creation and update of text + colors in a single call.

Gap 3: f_safeText — Added str.toString() Defensive Cast

Before:

```
f_safeText(string val) =>
  na(val) ? "-" : val
```

After:

```
f_safeText(string val) =>
  na(val) ? "-" : str.toString(val)
```

Rationale: `str.toString()` adds a defensive cast even though all current inputs are strings. If a future edit accidentally passes a float/int, `str.toString()` prevents a type error.

Additional Fixes Applied

Tracker Counters (v17.54.3 new)

```
var int sniperTrades    = 0    // Total sniper signals fired (long + short)
var int activeTrades    = 0    // Currently active trade count
var int totalAlertsFired = 0    // Total alerts fired across all types
```

- `sniperTrades` increments on `fire_sniper_long` and `fire_sniper_short`
- `totalAlertsFired` increments on ALL 4 fire events (sniper long/short + retrace long/short)
- `activeTrades` placeholder for future state machine integration
- All use `var` for persistence across chart refreshes

pdCryptoMode Toggle (v17.54.3 new)

```
pdCryptoMode = input.bool(false, "Crypto Mode", group=grpH4, tooltip="...")
```

- Wired into `pipSize` calculation: `pdCryptoMode or syminfo.type == "crypto" ? 1 : 10`
 - Default: `false` (Forex mode)
 - Group: H4 Multi-TF Bias settings
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Safety Patterns Summary

Pattern	Status	Evidence
<code>f_safeText()</code> on all dynamic values	✓	20 instances on all Col 1 data cells
<code>f_setCell()</code> wrappers on all HUD writes	✓	42 calls (0 bare <code>table.cell</code> in HUD)
<code>maxHudRows = 21</code> , <code>maxHudCols = 2</code>	✓	Exact fit with documented row map
Guard: <code>not na()</code> and <code>col ≥ 0</code> and <code>row ≥ 0</code> and <code>col < max</code> and <code>row < max</code>	✓	All 3 wrapper functions
Safe init: <code>var table hud = na +</code> <code>na(hud) guard</code>	✓	Lines 2749-2751
No global <code>max_bars_back</code>	✓	Removed in v17.54.2
<code>maxZoneAge = 500</code>	✓	Line 2168
<code>maxZones = 10 + FIFO</code>	✓	Line 2161, 6 <code>array.shift</code> calls
Tracker counters	✓	<code>sniperTrades</code> , <code>activeTrades</code> , <code>totalAlerts-Fired</code>
<code>pdCryptoMode toggle</code>	✓	Line 639, wired to <code>pipSize</code>

Risk Assessment

Risk	Level	Mitigation
HUD row overflow	LOW	maxHudRows=21 exact fit, bounds checked
na crash on dynamic text	LOW	All 20 dynamic values wrapped in f_safeText
table re-creation	LOW	<code>var table hud = na + na(hud) guard</code>
Tracker counter overflow	NEGLIGIBLE	Pine int supports $\pm 9.2 \times 10^{18}$
pdCryptoMode side effects	LOW	Only affects pipSize calc, default=false
H4/alert logic changes	NONE	No changes to request.security, alertcondition, bias thresholds, EQ calculations

What Was NOT Changed

- ❌ H4 P&D anchor logic (17 `request.security` calls unchanged)
- ❌ Alert conditions / alertcondition() messages (wording preserved)
- ❌ Bias thresholds / EQ calculations
- ❌ Alert wording or JSON payload structure
- ❌ Zone drawing logic
- ❌ Confluence scoring
- ❌ Signal hierarchy / deduplication

Artifacts Generated

1. `hotfix/hud-tracker-v17.54.3` branch — pushed to origin
2. `v17_54_3_unified.diff` — unified diff (5991 lines)
3. `V17_54_3_FIX_REPORT.md` — this report
4. `v17_54_3_VERIFICATION_REPORT.md` — verification template
5. Tag: `v17.54.3-hotfix-hud-tracker` (pending creation)

Next Steps

1. **User audit:** Review this report + unified diff
 2. **8-hour verification:** Run continuous 5M USDJPY test
 3. **30-min dense history test:** 5M & 1M scroll
 4. **Alert/tracker test:** Verify counter increments
 5. **Crypto toggle test:** Enable pdCryptoMode on BTC pair
 6. **User creates PR:** With proper description
 7. **Merge to main:** ONLY after user approval
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